

Let A be a dg algebra over a fixed base (commutative) ring Q .

Exercise 1. Show that if M is a dg A -module, then $H(M)$ is a graded $H(A)$ -module.

Exercise 2.* Decide if $R = k[[x]]/(x^2)$ and

$$M = \cdots \rightarrow R \xrightarrow{x} R \xrightarrow{x} R \xrightarrow{x} R \rightarrow \cdots$$

is a semiprojective dg R -module. Justify your answer completely.

Exercise 3. Let P be a dg A -module. Prove that P is semiprojective if and only if any/all of the following hold:

- (a) For any $M \xrightarrow{\sim} N$, the induced map $\text{Hom}_A(P, M) \rightarrow \text{Hom}_A(P, N)$ is a surjective quasiiso.
- (b) P is a projective graded A -module and $\text{Hom}_A(P, C) \simeq 0$ whenever $C \simeq 0$.
- (c) P is a direct summand of a semifree dg A -module.

Exercise 4. What dg A -modules are the projective objects in the category of dg A -modules?

Exercise 5.* The mapping cone of a semiprojective dg A -module is again semiprojective.

Exercise 6. Prove the following facts about semifree dg modules.

- (a) If F is semifree, then F is free as a graded A -module.
- (b) If A is a nonnegatively graded Q -algebra (i.e., $A_i = 0$ for $i < 0$) and F is a dg A -module that is bounded below and F is free as a graded A -module, then F is semifree over A .
- (c) If F is semifree, then F is semiprojective.

Exercise 7.* Let $Q = k[[x]]$ and $f = x^3$. Set $E = Q[e \mid \partial e = f]$ and consider the Q -complex M defined by:

$$0 \rightarrow Q \xrightarrow{\begin{bmatrix} 0 \\ -x \end{bmatrix}} Q^2 \xrightarrow{\begin{bmatrix} x & 0 \end{bmatrix}} Q \rightarrow 0$$

concentrated in homological degrees 0,1,2.

- (a) Check that the following maps prescribe M with distinct dg E -module structures:

$$0 \leftarrow Q \xleftarrow{\begin{bmatrix} 0 & -x^2 \end{bmatrix}} Q^2 \xleftarrow{\begin{bmatrix} x^2 \\ 0 \end{bmatrix}} Q \leftarrow 0 \quad \text{and} \quad 0 \leftarrow Q \xleftarrow{\begin{bmatrix} 1 & -x^2 \end{bmatrix}} Q^2 \xleftarrow{\begin{bmatrix} x^2 \\ 1 \end{bmatrix}} Q \leftarrow 0.$$

- (b) Are the dg A -modules from part (a) the same up to homotopy equivalence (or, equivalently, quasiisomorphism)?

Exercise 8.* Let $Q = k[[x, y]]$ and $f = x^3 + y^4$. Set $E = Q[e \mid \partial e = f]$ and $R = Q/(f)$. Consider the following complex F (concentrated in nonnegative degrees) that is periodic starting at homological degree 2.

$$\cdots \rightarrow Q^4 \xrightarrow{\begin{bmatrix} f & 0 & -y^3 & x^2 \\ 0 & f & x & y \\ -y & x^2 & 1 & 0 \\ x & y^3 & 0 & 1 \end{bmatrix}} Q^4 \xrightarrow{\begin{bmatrix} -1 & 0 & -y^3 & x^2 \\ 0 & -1 & x & y \\ -y & x^2 & -f & 0 \\ x & y^3 & 0 & -f \end{bmatrix}} Q^4 \xrightarrow{\begin{bmatrix} f & 0 & -y^3 & x^2 \\ 0 & f & x & y \\ -y & x^2 & 1 & 0 \\ x & y^3 & 0 & 1 \end{bmatrix}} Q^4 \xrightarrow{\begin{bmatrix} -y & x^2 & -f & 0 \\ x & y^3 & 0 & -f \\ 0 & -1 & x & y \end{bmatrix}} Q^3 \xrightarrow{\begin{bmatrix} x & y & f \end{bmatrix}} Q$$

- (a) Prove that F is a free Q -resolution of k .¹
- (b) Prove that the matrices below give F a dg E -module structure

$$\begin{aligned} F_1 &\xleftarrow{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}} F_0 & F_2 &\xleftarrow{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}} F_1 \\ F_{2n+1} &\xleftarrow{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}} F_{2n} & F_{2n+2} &\xleftarrow{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}} F_{2n+1}, \quad \text{for } n \geq 1. \end{aligned}$$

- (c) Prove that F , equipped with the dg E -module structure from part (b), is a semifree resolution of k over E .
- (d) Prove that $F \otimes_E R$ is the minimal free resolution of k over R .

¹For those skeptical, check this is correct with `Macaulay2`.